C

ACT SUBJECT TEST — MATH 1 • UNIT 0.01

ACT II Math Level 1 Diagnostic & Strategy Drill

PRACTICE WORKSHEET

Algebra & Geometry Readiness • Official ACT Subject Test Curriculum

Name: _____ **Date:** _____ **Score:** _____ / 25

ACT II Math Level 1 Diagnostic & Strategy Drill Practice Exercises

1 MCQ

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

A 1

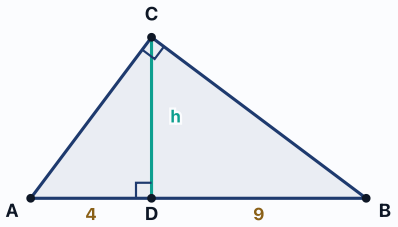
B 3

C 4

D 5

2 MCQ

In right triangle ABC, altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of altitude $\overline{BD}$?



A 6

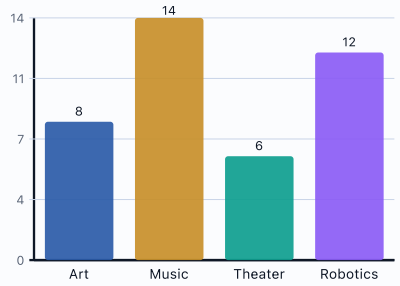
B 6.5

C 13

D 36

3 MCQ

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?



A $\frac{7}{50}$

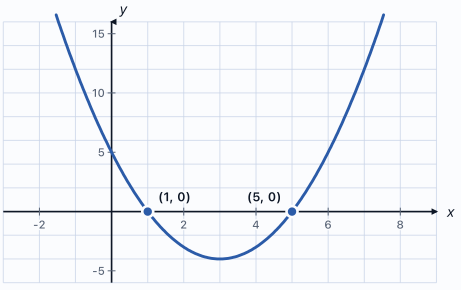
B $\frac{7}{20}$

C $\frac{7}{13}$

D $\frac{13}{20}$

4 MCQ

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



A 12

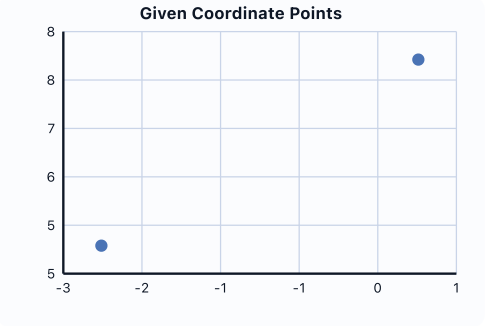
B 8

C 4

D -3

5 MCQ

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?



A 25

B 5

C 7

D $\sqrt{13}$

6 MCQ

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

A 5

B 11

C 6

D 8

7 MCQ

If $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, what is the value of $g(f(2))$?

A 9

B 12

C 13

D 21

8 MCQ

What is the smallest positive integer n such that $126n$ is the square of an integer?

A 7

B 14

C 21

D 42

9 MCQ

For a positive integer k, the greatest common divisor of k and 24 is 6, and the least common multiple of k and 40 is 600. What is the value of k?

A 120

B 300

C 150

D 75

10 MCQ

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

A 34

B 80

C 128

D 64

11 MCQ

In the standard (x, y) coordinate plane, the system of equations

$$\begin{cases} y = |x^2 - 4| \\ y = 2x + k \end{cases}$$

has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

A 5

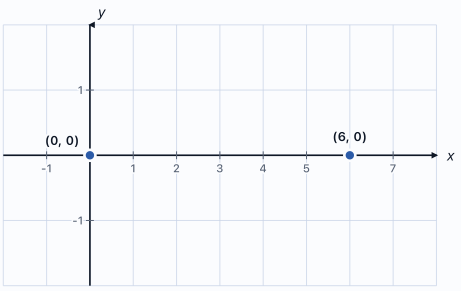
B 9

C 11

D 4

12 MCQ

In the standard (x, y) coordinate plane, the vertices of $\triangle OAB$ are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides $\triangle OAB$ into two regions of equal area. What is the value of k?



A $6 - 2\sqrt{3}$

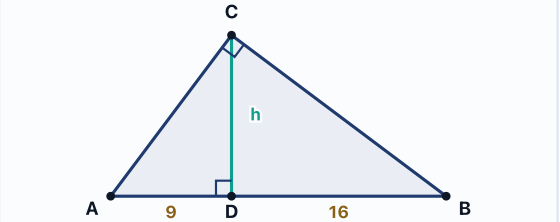
B $\sqrt{6}$

C $\frac{8}{3}$

D 3

13 MCQ

In the figure below, $\triangle ABC$ is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in $\triangle ACD$, and a circle with center O_2 is inscribed in $\triangle BCD$. What is the distance between the centers O_1 and O_2?



A 5

B 7

C $7\sqrt{2}$

D $5\sqrt{2}$

14 MCQ

Consider the system of linear equations below:

$$\begin{cases} 3x - 2y = 8 \\ kx + 6y = 5 \end{cases}$$

For which of the following values of the constant k does the system have no solution (x, y)?

A 4

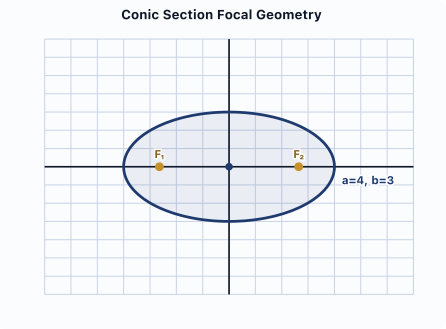
B 9

C -4

D -9

15 MCQ

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49} + \frac{(y+2)^2}{16} = 1$?



A 7

B 8

C 14

D 49

16 MCQ

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

A 8

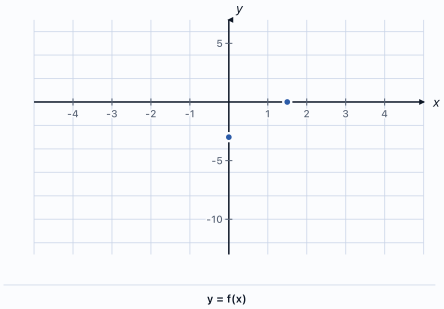
B 10

C 9

D 12

17 MCQ

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?



A -7

B 12

C 15

D 9

18 MCQ

In the standard (x, y) coordinate plane, a right triangle has vertices at $A(-3, -1)$, $B(5, 5)$, and $C(4, k)$, where the right angle is at vertex C and $k > 0$. What is the value of k?

A 8

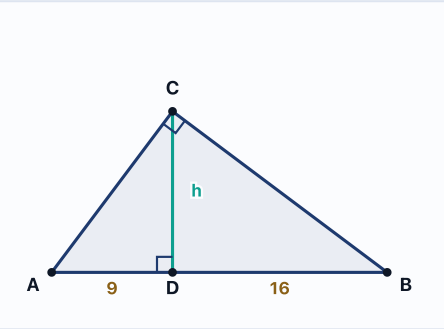
B 2

C 4

D 6

19 MCQ

In the figure below, $\triangle ABC$ is a right triangle with right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$ with length h. If $AD = 9$ and $DC = 16$, what is the perimeter of $\triangle ABC$?



A 47

B 60

C 50

D 72

20 MCQ

If the real numbers x and y satisfy the system of equations

$$x^3 - 3xy^2 = 10$$
$$3x^2y - y^3 = 5$$

what is the value of $x^2 + y^2$?

A 25

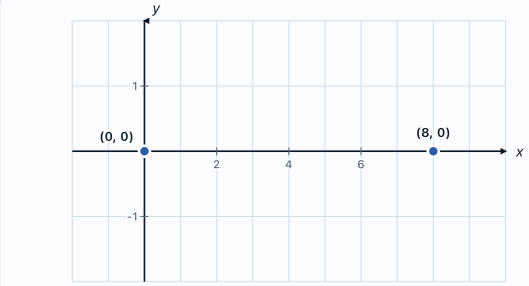
B $\sqrt{5}$

C $5\sqrt{5}$

D 5

21 MCQ

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?



A 4

B $\frac{10}{3}$

C $2\sqrt{6}$

D $8 - 2\sqrt{6}$

22 MCQ

In $\triangle ABC$, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides $\triangle ABC$ into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

A $\frac{7\sqrt{2}}{2}$

B $4\sqrt{3}$

C $2\sqrt{6}$

D $6\sqrt{2}$

23 MCQ

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee?

A $\frac{3}{7}$

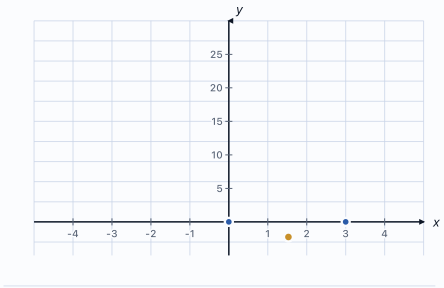
B $\frac{1}{3}$

C $\frac{9}{35}$

D $\frac{3}{5}$

24 MCQ

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?



A $x = -2$ and $x = 2$

B $x = 2$ and $x = 3$

C $x = -4$ and $x = -1$

D $x = 1$ and $x = 4$

25 MCQ

In the standard (x, y) coordinate plane, what are the center and the radius of the circle with equation $(x - 4)^2 + (y + 6)^2 = 49$?

A Center $(4, -6)$, radius 7

B Center $(-4, 6)$, radius 7

C Center $(4, -6)$, radius 49

D Center $(-4, 6)$, radius 49